Muhammad Ehsan

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ABOUT MYSELF

Artificial Intelligence Engineer with a strong foundation in **Deep Learning** and **Full-stack Software Engineering**. Proven track record in developing **real-time computer vision systems** using YOLOv8/v11, specializing in **instance segmentation** and predictive analytics. Experienced in building robust**backend architectures**, API integrations, and automated testing pipelines. A **published researcher** committed to deploying scalable, **intelligent systems** that solve complex urban and biomedical challenges.

WORK EXPERIENCE

Chaoyang University of Technology Taichung University of Technology, Taiwan

University research assistant

01/11/2024 - Current

- **Developed and deployed real-time computer vision systems** using YOLOv8/v11 for high-accuracy object detection and instance segmentation in urban environments.
- **Engineered end-to-end deep learning pipelines** in PyTorch and TensorFlow, managing the full lifecycle from data curation to hyperparameter tuning.
- **Optimized model inference for edge applications** using OpenCV, focusing on stream handling and latency reduction for real-world deployment.
- **Architected automated Python workflows** for predictive analytics and data integration, significantly enhancing research efficiency and backend reliability.
- **Collaborated on applied R&D projects**, bridging deep learning theory with scalable system development for smart city and biomedical applications.

Engin Technologies Lahore, Pakistan

Software engineer

01/02/2023 - 30/09/2024

- **Scaled backend systems** to ensure high availability and performance for core features.
- **Built RESTful APIs** and integrated third-party services for data synchronization.
- **Automated CI/CD & testing** to streamline deployments and ensure code reliability.
- **Refactored legacy code** to reduce technical debt and improve system efficiency.
- **Designed secure architectures** meeting modern scalability and performance standards.

Freelance Lahore, Pakistan

artificial intelligence engineer

01/02/2024 - 30/09/2024

- **Architected and deployed end-to-end automation** by integrating RPA workflows with custom AI and Machine Learning models.
- **Developed Python-based computer vision tools** for automated data extraction and processing from unstructured sources.
- **Consulted for diverse clients to identify bottlenecks** and implement scalable automation, significantly reducing manual workloads.
- **Integrated third-party APIs and data pipelines**, ensuring seamless

communication between automation bots and backend systems.

EDUCATION & TRAINING

01/10/2024 - CURRENT Taichung City, Taiwan

Master of Science in Information Management Chaoyang University of Technology

Level in EQF: EQF level 7 | **Thesis:** Smart Parking Validation using YOLO Segmentation and Geometric Analysis.

01/09/2019 - 31/07/2023 Lahore, Pakistan

Bachelor of Science in Computer Science University of Central Punjab

Level in EQF: EQF level 6 | **Thesis:** Prophecy – A Blockchain Based Decentralized Prediction Platform

PUBLICATIONS

Enhancing Learning Through Summarization Using a Fine-Tuned T5 Model

Ehsan, M., et al., 2026, Lecture Notes in Computer Science (LNCS), vol 15914, pp. 269-278.

Smart Detection of Open Parking Spaces in Urban Taiwan Using YOLOv8 Segmentation on Dashcam Footage

Ehsan, M., Chen, RC., 2026, IIHMSP 2025, Springer. (In Press).

PROJECTS

Cytokine Detection Using Deep Learning, Samsung Innovation Program (SIC)

- **Developed CNN-based models & biomarker pipelines** for automated cytokine detection and immune response analysis to assist in early disease diagnosis.
- **Optimized accuracy & scalability** for high-stakes biomedical datasets, bridging deep learning theory with practical healthcare automation and diagnostic tools.

SKILLS

ARTIFICIAL INTELLIGENCE & DEEP LEARNING

Python | CI/CD Pipelines | Machine Learning (AI, DL, CNN) | Instance Segmentation | Object Detection | AI platforms(PyTorch, Tensorflow, TensorRT, ONNX Runtime) | Python & libraries (Numpy, Matplotlib, Scikit-learn, Pandas, Pytorch) | Image Preprocessing and Image Classification | AI Data Annotation | Frameworks and Libraries: FastAPI, Open CV, Tensorflow, Keras, Pandas, Seaborn, Plotly, Streamlit)

SOFTWARE ENGINEERING & BACKEND

Ruby On Rails | C, C++,C# | RESTful APIS | Git/Github, Docker, Gitlab | Microservices | System Architectures | Unit Testing | Integration Testing | Automated Testing

DATA ENGINEERING & IOT

Data Pipelines | AWS | Databases (MongoDB MySql PostgresSql) | MQTT | ESP32 | Sensor Data Integration

AUTOMATION (RPA)

Robotic Process Automation (RPA). | Python Scripting | UiPath | Power Automate

CERTIFICATIONS

Samsung Innovation Campus (SIC), 03/12/2022

Artificial Intelligence

iNeuron, 19/01/2024

Foundational Generative Ai

LANGUAGE SKILLS

MOTHER TONGUE(S): Urdu
OTHER LANGUAGE(S): English | Chinese